# Lakshmi Lahari Satti

## AI/ML Engineer | 5 Years

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### Summary

**AI/ML Engineer** with nearly **5 years of experience** building machine learning and LLM-based systems across research and production environments. Work includes data preparation pipelines, feature engineering, model training, evaluation workflows, and inference services supporting prediction tasks. Experience with RAG pipelines, embeddings, experiment tracking, and monitoring processes across cloud platforms. Comfortable collaborating with data scientists and product teams to refine datasets, review model behavior, and maintain reliable performance in deployed machine learning systems.

### Technical Skills

- **Programming & Data:** Python, R, SQL, PostgreSQL, MySQL, Pandas, NumPy, SciPy
- **Machine Learning:** Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning, A/B Testing, Error Analysis, Class Imbalance Handling
- **Deep Learning & NLP:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Transformer Architectures, Text Classification, Named Entity Recognition, Sentiment Analysis, Sequence Modeling
- **Generative AI & LLMs:** Prompt Engineering, Retrieval-Augmented Generation (RAG), Embeddings, Vector Search, FAISS, Pinecone, OpenAI API, Azure OpenAI, LoRA, QLoRA, PEFT, LangChain, LlamaIndex, Hallucination Evaluation
- **Data Engineering & Pipelines:** Data Preprocessing, ETL Workflows, Feature Pipelines, Structured Data Handling, Unstructured Data Processing, Dataset Versioning, Data Validation, Batch Inference, Real-Time Inference
- **MLOps & Lifecycle:** MLflow, Experiment Tracking, Model Versioning, CI/CD for ML, Docker, Kubernetes, Model Serving, Drift Detection, Inference Monitoring, Retraining Workflows, Logging
- **Deployment & Integration:** REST APIs, FastAPI, Flask, Backend Integration, API Integration, Microservices
- **Cloud Platforms:** AWS, Azure, GCP
- **Tools & Collaboration:** Git, Linux, Jira, Agile/Scrum, Technical Documentation

### Professional Experience

**AI/ML Engineer | *Scale AI*** **July 2025 – Present | *USA***

- Large annotation datasets structured for machine learning training workflows, preparing organized inputs for computer vision and language model systems, expanding dataset preparation throughput by 32% across active model development cycles.
- Python preprocessing pipelines converted raw multi-source records into standardized training datasets for classification experiments, cutting dataset preparation time by 29% while maintaining consistent feature formatting across project environments.
- Validation metrics and error diagnostics examined during model testing phases, allowing research teams to adjust training strategies and lifting classification performance by 25% across iterative development cycles.
- REST-based inference endpoints deployed to support internal prediction requests, allowing production models to process automated workloads while lowering average inference latency by 37% during system evaluation benchmarks.
- Annotation reviews conducted with data scientists to correct labeling inconsistencies within training datasets, reducing annotation noise and mislabeled sample rates by 28% during supervised learning preparation.
- Monitoring dashboards configured to observe prediction patterns, dataset drift signals, and evaluation statistics, helping engineering teams maintain stable model behavior and enhancing production model reliability by 33%.

**AI/ML Engineer | *Heptic AI*** **Jan 2020 – Dec 2023 | *India***

- Training datasets collected from relational databases and external data sources, organizing labeled records used in classification and prediction experiments while streamlining dataset preparation efficiency by 27% across research projects.
- Python notebooks used to explore dataset distributions, detect anomalies, and examine variable relationships prior to model development, cutting preprocessing errors by 32% during data preparation stages.
- Supervised learning models trained using prepared datasets while validation testing compared predicted outputs with labeled records, raising prediction accuracy by 30% across experimental model iterations.
- Feature engineering procedures applied to transform raw variables through encoding and normalization methods, stabilizing model inputs and lowering feature inconsistencies by 26% across machine learning experiments.
- Batch inference scripts executed trained models to generate prediction outputs for analytics teams, enabling automated analysis workflows and improving processing efficiency by 28% across internal reporting pipelines.
- Experiment tracking records maintained to document training configurations and evaluation results, enabling reproducible model experiments and strengthening experiment traceability by 22% across development environments.

### Projects

**Intelligent Document Classification and Sensitive Data Detection System**

- Labeled document datasets prepared for supervised learning experiments while Python preprocessing pipelines cleaned raw text, normalized language content, and organized structured inputs suitable for transformer-based classification models.
- Transformer-based models trained on prepared datasets while evaluation metrics compared predicted labels with validation samples to examine classification behavior during iterative development cycles.
- Embedding vectors generated from processed documents enabling vector search across document collections, while a RAG retrieval workflow connected semantic search results with inference endpoints supporting automated classification requests.

### Education

**Master's in Information Science (Machine Learning)** **Jan 2024 - Dec 2025| *USA***

University of Arizona — Tucson, AZ